

SHUBHAM GAJJAR

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EDUCATION

Northeastern University, Portland, Maine
Master of Science in Artificial Intelligence

September 2025 – May 2027

LDRP Institute of Technology and Research, Gandhinagar, India
Bachelor of Engineering in Computer Engineering, Grade Point Average: 8.41/10.0

September 2022 – May 2025

VPMP Polytechnic, Gandhinagar, India
Diploma in Computer Engineering, Grade Point Average: 9.22/10.0

September 2019 – May 2022

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures and Algorithms, Image Processing

TECHNICAL SKILLS & CERTIFICATION

Programming Languages: Python, JavaScript

Deep Learning: TensorFlow, Keras, PyTorch, CUDA

Computer Vision: OpenCV, Matplotlib, Albumentations

Data Science: NumPy, Pandas, Jupyter, Scikit-learn, Seaborn

Web Development: Next.js, React, JavaScript, Tailwind CSS, Flask

Tools: Git, Vercel, Framer Motion, Typst

Certificates: Python for Data Science from Indian Institute of Technology Madras, Python Data Structures from University of Michigan

PROFESSIONAL EXPERIENCE

IDEXX Laboratories & Northeastern Institute for Experiential AI, Portland, Maine

Research Assistant - Vision Language Models

March 2026 – Present

- Developing Vision Language Models using PyTorch for cellular imaging analysis in IDEXX's inVue Dx Cellular Analyzer, generating pathologically-informed structured text descriptions for clinical deployment in veterinary diagnostics
- Implementing multimodal architectures combining computer vision encoders with language models to produce structured, clinically-accurate diagnostic outputs from biomedical scanner images
- Collaborating with cross-functional team including Dr. Michael Wan (Research Scientist, EAI), David Steines (Sr. Director of Machine Intelligence, IDEXX), IDEXX AI engineers, and veterinary pathologists to align model outputs with clinical requirements

BigCircle (UPSAAS Technologies LLP), Gandhinagar, India

Artificial Intelligence Engineer

January 2025 – August 2025

- Developed end-to-end Deep Research pipeline using Python orchestrating ML prompt generation, Firecrawl API for web scraping, ChatGPT API summarization, graph visualization with Matplotlib and Seaborn, and automated Typst PDF reports; optimized API call efficiency reducing processing time by 75%
- Engineered pagination and authentication systems using JavaScript and Next.js, accelerating page load times by 40% with Docker containerization for 500+ concurrent sessions
- Collaborated with cross-functional teams including product managers and non-technical stakeholders to deliver iOS applications using React Native
- Worked in 5-member Agile team using Git for version control; performed code reviews to improve code quality and maintainability

KEY RESEARCH PROJECTS

Hybrid ResNet-ViT for Skin Cancer Classification

Published in IEEE, 2025

- Hybrid architecture using TensorFlow combining frozen ResNet50 with Vision Transformer blocks, achieving 96.3% accuracy on HAM10000 dataset
- Integrated multi-head self-attention for seven-class classification with Area Under Curve of 1.00; visualized performance using Matplotlib
- Presented research at IEEE World Conference on Applied Intelligence and Informatics

Extended ResNet50 with Inverse Soft Mask Attention for Skin Cancer

Submitted to journal, 2025

- Two-stage pipeline using TensorFlow and Keras with U-Net++ hair segmentation and Extended ResNet50 classifier, achieving 97.89% accuracy on HAM10000 dataset
- Integrated dense residual blocks and Squeeze-and-Excitation modules with weighted feature aggregation
- Conducted 21 architectural trials with custom attention mechanisms outperforming baseline methods

VGG16-MCA UNet for Brain Tumor Segmentation

Under Review at Elsevier

- VGG16-based encoder with Multi-Channel Attention decoder using TensorFlow, achieving 99.59% accuracy on LGG Brain MRI dataset
- Ensemble learning combining multiple Keras configurations improved Dice coefficient by 3.7% over standard UNet
- Preprocessing pipeline with NumPy and Pandas implementing skull stripping and intensity normalization using Scikit-learn